ACCRA TECHNICAL UNIVERSITY

Department of Computer Science

BCS 404: Introduction to Data Science with Python

PROJECT WORK

Project Title:

Exploratory Data Analysis, Statistical Analysis and Machine Learning using a Real-World Dataset

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1. Introduction

The purpose of this project is to demonstrate practical competence in applying Python libraries for data visualization, statistical analysis, and introductory machine learning on a real-world dataset. Using the widely recognised Titanic dataset, this study conducts exploratory data analysis to uncover patterns in passenger demographics and survival rates. Furthermore, a classification model is developed to predict passenger survival based on selected features, applying techniques to clean, analyze, and model the data using tools such as Pandas, Matplotlib, Seaborn, and Scikit-Learn.

2. Dataset Description

The dataset used for this analysis is the Titanic Dataset, downloaded directly from Kaggle. It contains records of passengers aboard the RMS Titanic, detailing various attributes such as age, gender, passenger class (Pclass), ticket fare, port of embarkation, and whether they survived the disaster. Initially, the data contains continuous, categorical, and text-based variables, some of which exhibit missing or null values that require careful preprocessing before statistical or predictive modeling can occur.

3. Methodology

The data was first imported into a Jupyter Notebook environment using the Pandas library. To prepare the data for analysis, specific data cleaning operations were executed. Missing values were detected and handled appropriately. The Age column's missing values were replaced with the median age to minimize the influence of outliers. Missing values in the Embarked column were filled using the mode, as it is a categorical variable. The Cabin column was dropped entirely due to a high volume of missing observations that rendered it unusable. The dataset was also checked for duplicated observations, which were subsequently removed to ensure data integrity.

For the machine learning phase, categorical variables were transformed into numerical format using dummy variables. Non-predictive text columns (such as Name and Ticket) were dropped. The dataset was then split into independent variables and the target variable (Survived), and divided into training and testing sets. A Logistic Regression classifier was trained on the training set to predict survival on the testing set.

4. Results

Descriptive statistics and frequency distributions revealed that the majority of passengers did not survive the sinking. The passenger base was heavily concentrated in the third class and the median age hovered around young adulthood.

Correlation analysis produced several key insights. The strongest positive correlation identified was between the SibSp (siblings/spouses) and Parch (parents/children) variables, yielding a value of roughly 0.41. The strongest negative correlation was discovered between Pclass and Fare at approximately -0.55.

The Logistic Regression model was used to predict testing data. The resulting model yielded an accuracy of approximately 80%. The generated Confusion Matrix and Classification Report indicated that the model possessed a solid ability to classify outcomes, though it exhibited slightly higher precision and recall when predicting non-survivors compared to survivors.

5. Discussion

The statistical insights support the historical narratives of the Titanic disaster. The strong negative correlation between Pclass and Fare logically confirms that higher-tier accommodations required costlier tickets. Visually, the pairplots and bar charts highlighted that passengers in first class, as well as female passengers, had notably higher survival rates. The strong positive correlation between SibSp and Parch indicates that passengers largely traveled in family units rather than alone.

The machine learning results demonstrate that Logistic Regression serves as an effective, efficient baseline model for this binary classification problem, achieving roughly 80% accuracy. However, there are limitations to this study. Dropping the Cabin column, while necessary due to missing data, removed potentially valuable insights regarding a passenger's physical location on the ship. Additionally, imputing missing ages with the median may have slightly distorted the true age variance.

6. Conclusion

This project successfully implemented an end-to-end data science workflow, transforming raw data into actionable statistical insights and a functional predictive model. The exploratory analysis confirmed that survival on the Titanic was heavily influenced by passenger class, gender, and ticket fare. For future iterations, it is recommended to apply more complex machine learning algorithms, such as Random Forests or Gradient Boosting, and to engineer new features from the text data (like extracting passenger titles) to increase the model's predictive accuracy potentially.

7. References

- Kaggle. (n.d.). *Titanic Dataset*. Retrieved from https://www.kaggle.com/competitions/titanic/data
- Dadzie, J. (2026). *BCS 404: Introduction to Data Science with Python - Project Work*. Accra Technical University, Department of Computer Science.

Appendix (Python Code)

```python
# Task 1: Data Acquisition
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv('titanic.csv')
print("Dataset Dimensions:", df.shape)
print("\nColumn Names:", df.columns.tolist())
display(df.head())
print("\nData Types:")
print(df.dtypes)

# Task 2: Data Cleaning
df['Age'] = df['Age'].fillna(df['Age'].median())
df['Embarked'] = df['Embarked'].fillna(df['Embarked'].mode()[0])
df = df.drop(columns=['Cabin'])
if df.duplicated().sum() > 0:
    df = df.drop_duplicates()
```

```python
# Task 3: Data Visualisation
sns.set_theme(style="whitegrid")

plt.figure(figsize=(8, 5))
sns.histplot(df['Age'], bins=30, kde=True, color='skyblue')
plt.title('Histogram of Passenger Ages')
plt.xlabel('Age')
plt.ylabel('Frequency')
plt.show()

plt.figure(figsize=(8, 5))
sns.countplot(data=df, x='Pclass', palette='Set2')
plt.title('Passenger Class Distribution')
plt.xlabel('Passenger Class')
plt.ylabel('Count')
plt.show()

plt.figure(figsize=(8, 5))
sns.boxplot(data=df, x='Pclass', y='Age', palette='Set3')
plt.title('Boxplot of Age by Passenger Class')
plt.xlabel('Passenger Class')
plt.ylabel('Age')
plt.show()

plt.figure(figsize=(8, 5))
sns.scatterplot(data=df, x='Age', y='Fare', alpha=0.6, color='purple')
```

```python
plt.title('Scatter Plot of Age vs. Fare')
plt.xlabel('Age')
plt.ylabel('Fare')
plt.show()

plt.figure(figsize=(8, 6))
numeric_df = df.select_dtypes(include=['number'])
sns.heatmap(numeric_df.corr(), annot=True, cmap='coolwarm', fmt=".2f")
plt.title('Correlation Heatmap of Numerical Variables')
plt.show()

selected_vars = df[['Age', 'Fare', 'Pclass', 'Survived']]
sns.pairplot(selected_vars, hue='Survived', palette='husl')
plt.suptitle('Pairplot of Selected Numerical Variables', y=1.02)
plt.show()

# Task 4: Statistical Analysis
print(df.describe())
print(df['Survived'].value_counts())
corr_matrix = numeric_df.corr()
corr_pairs = corr_matrix.unstack().sort_values(ascending=False)
corr_pairs = corr_pairs[corr_pairs != 1.0]
print("Strongest Positive Correlation:\n", corr_pairs.head(2))
print("Strongest Negative Correlation:\n", corr_pairs.tail(2))
```

```python
# Task 5: Machine Learning
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

df_ml = pd.get_dummies(df, columns=['Sex', 'Embarked'], drop_first=True)
X = df_ml.drop(columns=['Survived', 'PassengerId', 'Name', 'Ticket'])
y = df_ml['Survived']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_sta

model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred))
print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred))
print("Classification Report:\n", classification_report(y_test, y_pred))
```